

CS-112

Object Oriented Programming

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Topics To be Covered Today

☐ Other OOP concepts



CLO Covered

CLO1: Understand the underlying concepts of object-oriented paradigm.

OOP Concepts

Some other terms which are used in Object-Oriented design:

- ❑ Association
- ❑ Aggregation
- ❑ Composition

Association

Association

- It represents the relationship between objects.
- It defines how objects interact and work together to achieve a common goal.

For example

Consider a **Teacher** and **Student** in a school system. A teacher can teach multiple students, and a student can learn from multiple teachers.

Here, **Teacher** and **Student** have an association where they interact but exist independently. This relationship can be **one-to-one**, **one-to-many**, or **many-to-many**, depending on the use case.

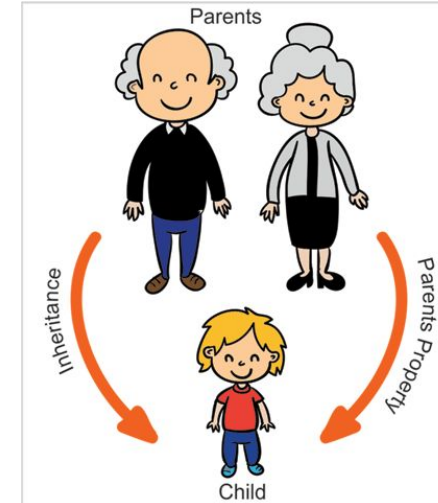
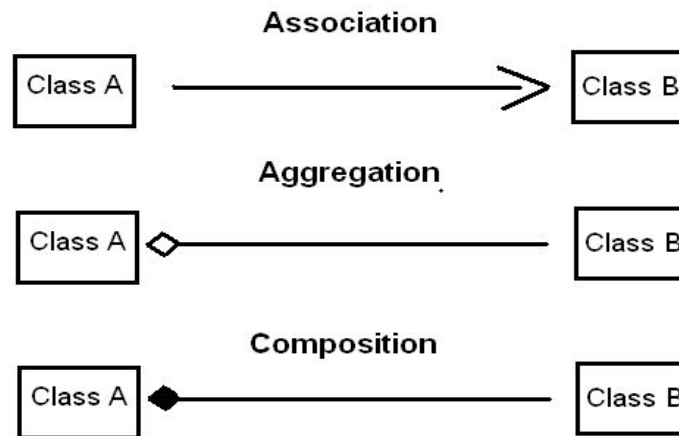
Kinds of Association

Class Association

- Inheritance

Object Association

- Simple Association
- Composition
- Aggregation



Simple Association

Is the weakest link between objects

Is a reference by which one object can interact with some other object

Is simply called as “association”

Kinds of Simple Association

w.r.t navigation (Direction of relationship)

- One-way Association (unidirectional)
- Two-way Association (bidirectional)

w.r.t number of objects

- Binary Association
- Ternary Association
- N-ary Association

One-way Association

One object knows about and interacts with another object, but the reverse is not true.

We can navigate along a single direction only.

Denoted by an arrow towards the server object.

Example – Association



Ali lives in a House

Example – Association



Ali drives his Car

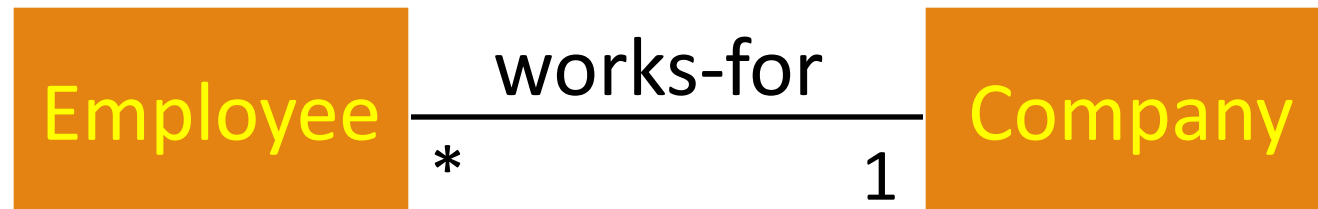
Two-way Association

Both objects are aware of each other and can interact.

We can navigate in both directions

Denoted by a line between the associated objects.

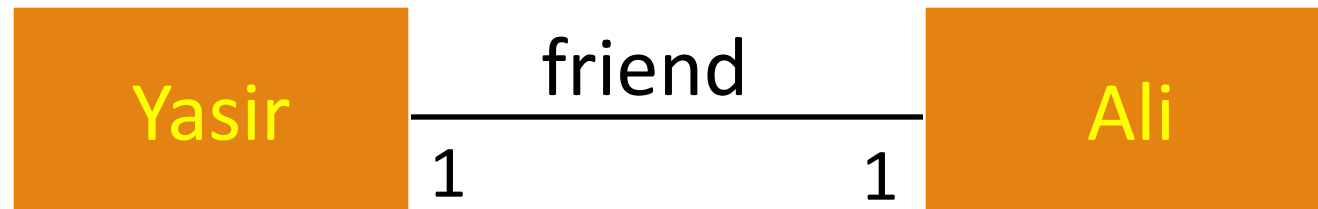
Example – Two-way Association



Employee works for company

Company employs employees

Example – Two-way Association



Yasir is a friend of Ali

Ali is a friend of Yasir

Kinds of Simple Association

W.R.T TO NUMBER
OF OBJECTS

Binary Association

Associates objects of exactly two classes

Denoted by a line, or an arrow between the associated objects

Example – Binary Association



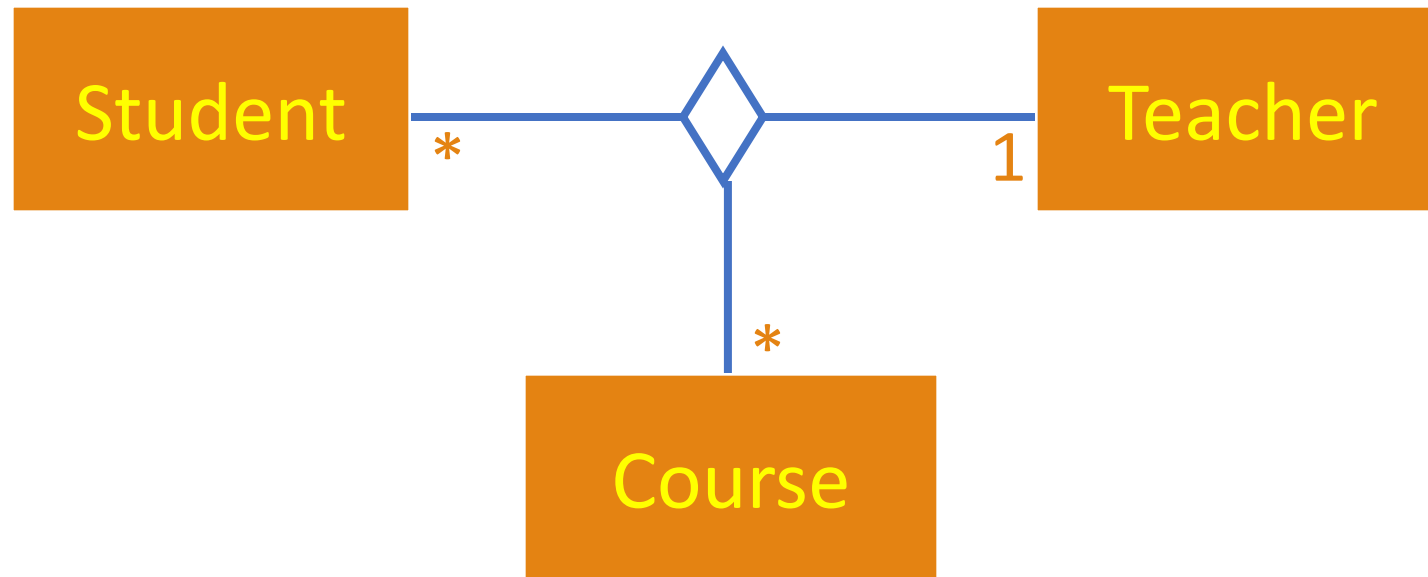
Association “drives” associates objects of exactly two classes

Ternary Association

Associates objects of exactly three classes

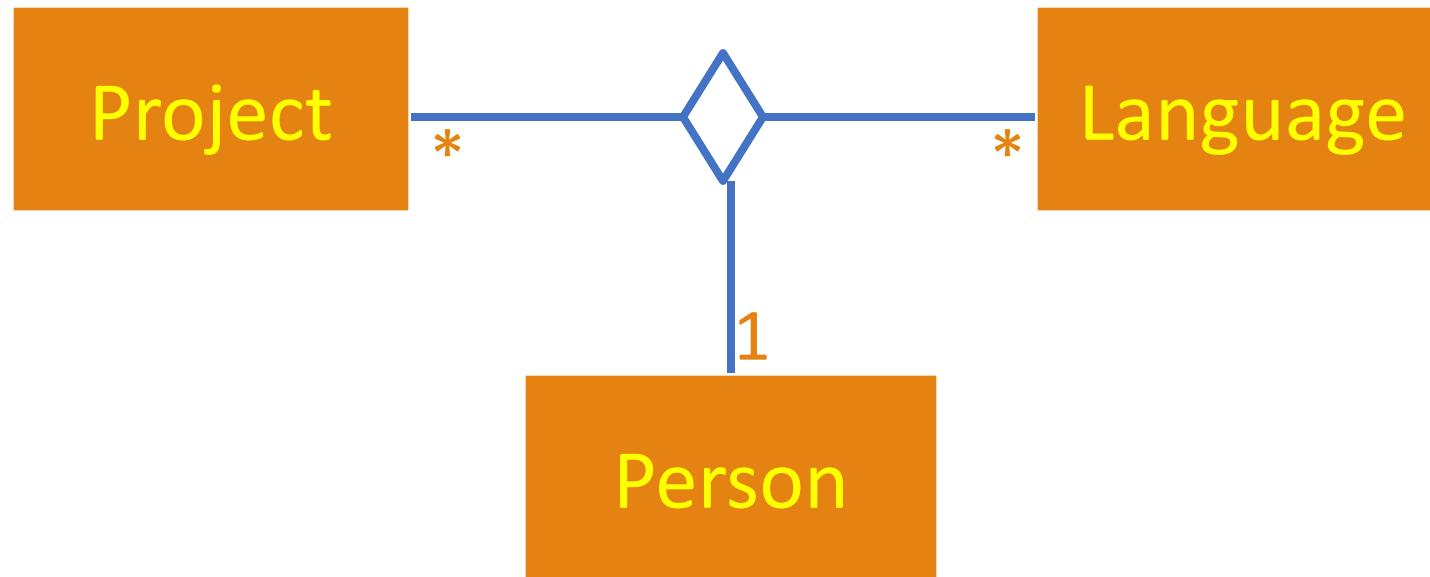
Denoted by a diamond with lines connected to associated objects

Example – Ternary Association



Objects of exactly three classes are associated

Example – Ternary Association



Objects of exactly three classes are associated

N-ary Association

An association between 3 or more classes

Practical examples are very rare

Example: A **Research Team** consists of multiple **Scientists**, working on multiple **Projects**, funded by multiple **Organizations**

Here, multiple **Scientists**, multiple **Projects**, and multiple **Organizations** are connected under a **Research Team**, forming an **N-ary Association**.

Composition



Composition

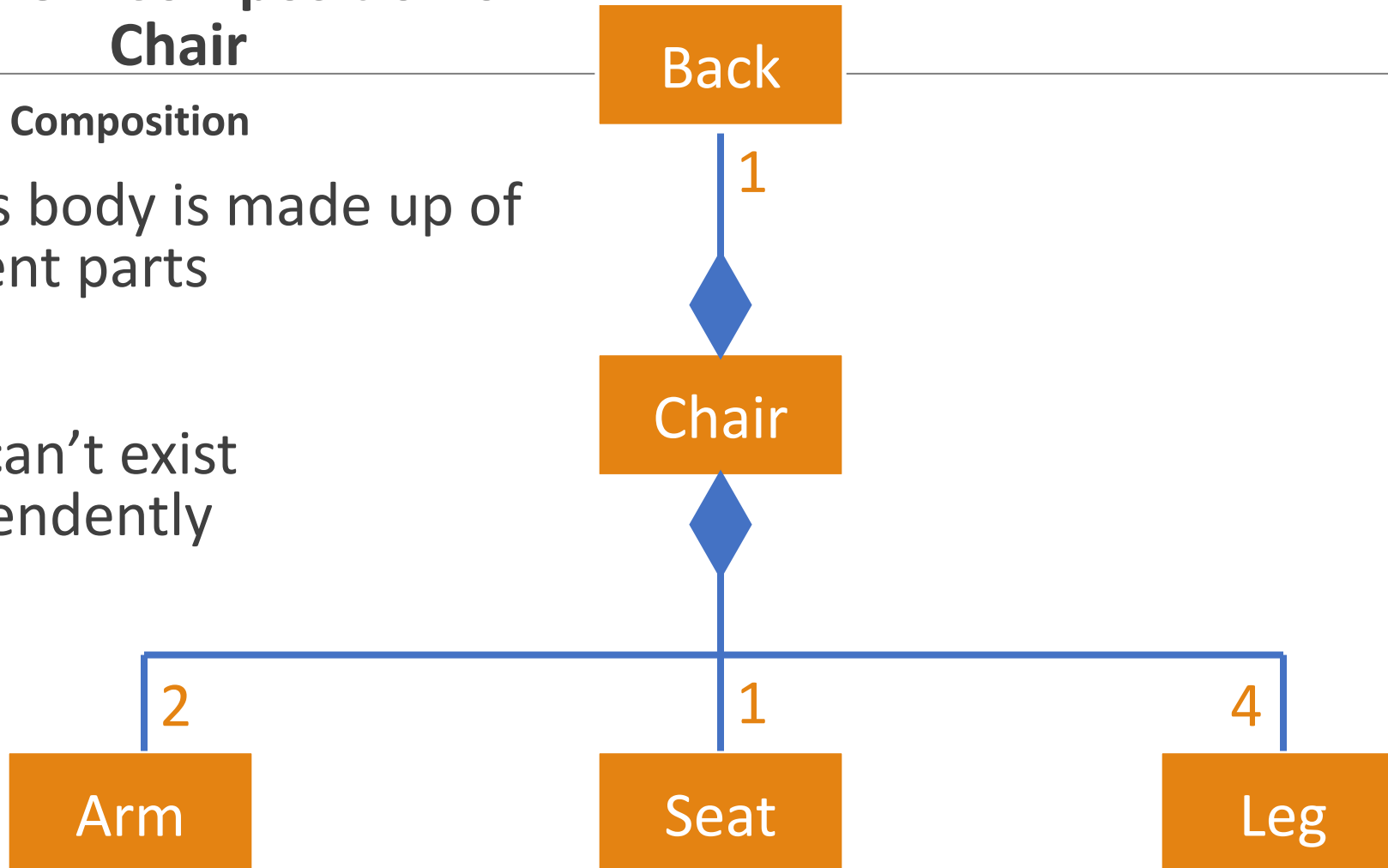
- An object may be composed of other smaller objects
- The composition represents the relationship where one object contains other objects as a part of its state.
- There is a strong relationship between the containing(parent) object and the dependent (child) object.
- It is the state where containing objects **do not have an independent existence**. If you delete the parent object, all the child objects will be deleted automatically.
- Composition is represented by a line with a filled-diamond head towards the parent object

Example – Composition of Chair

Stronger Composition

Chair's body is made up of different parts

They can't exist independently



Example – Composition is Stronger

Ali is made up of different body parts

They can't exist independent of Ali

A car has an engine. The engine cannot exist separately from the car.
If the car is destroyed, the engine is also destroyed.

A house has rooms. The rooms depend on the house's existence.

Aggregation

Aggregation

An object may contain a collection (aggregate) of other objects

The relationship between the container(Parent) and the contained (Child) object is called aggregation

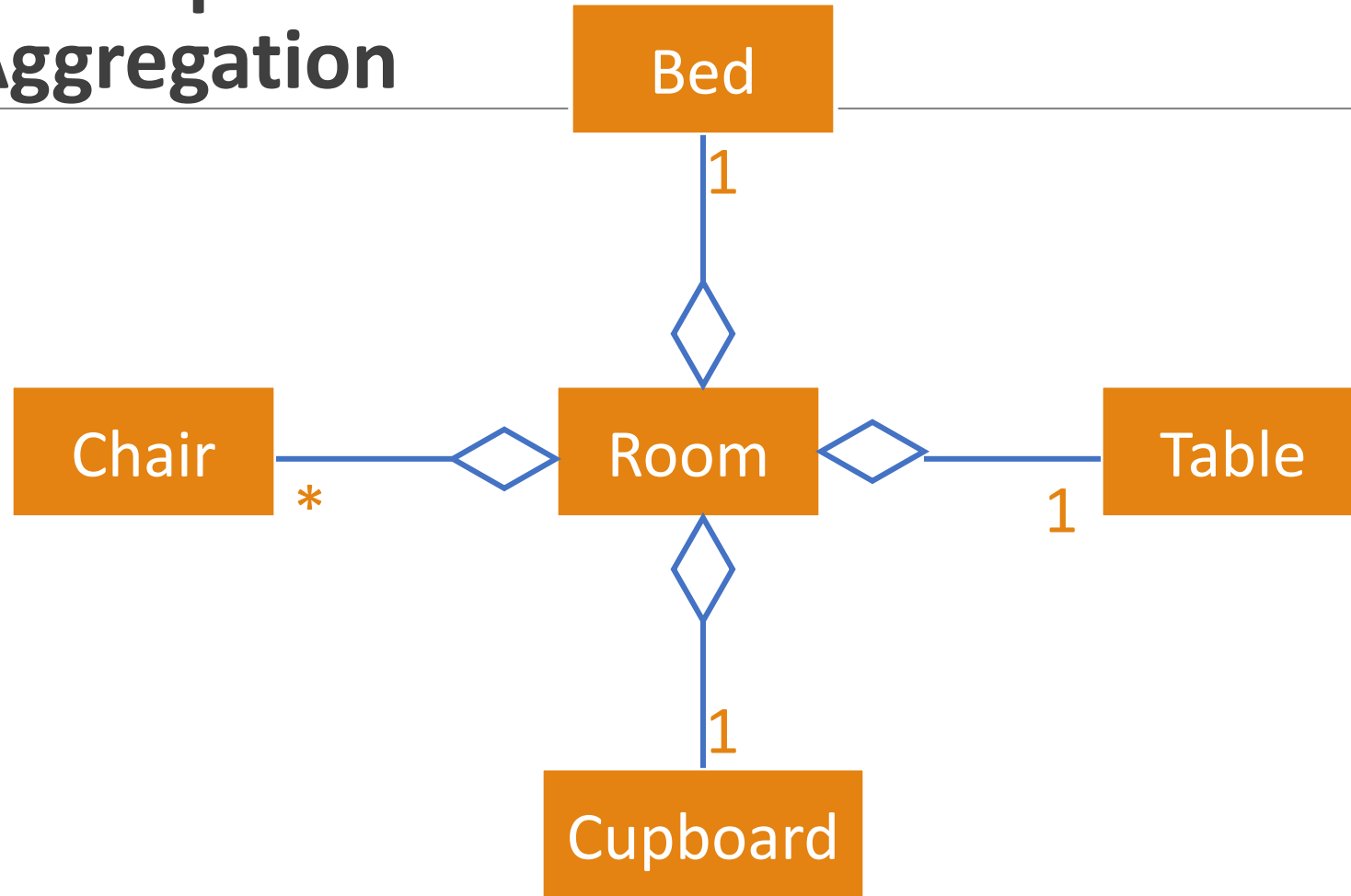
Aggregation is represented by a line with unfilled-diamond head towards the container (Parent Class)

Aggregation is weaker relationship, because

- Aggregate object is not a part of the container(Parent Class)

- Aggregate object can exist independently

Example – Aggregation



Example – Aggregation is Weaker

Furniture is not an intrinsic part of room

Furniture can be shifted to another room, and so can exist independent of a particular room

Example – Aggregation is Weaker

A plant is not an intrinsic part of a garden

It can be planted in some other garden, and so can exist independent of a particular garden